

4.11 Tailstock Setup and Operation

The ST-10 tailstock is manually positioned, then the quill is hydraulically applied to the workpiece. Command hydraulic quill motion using the following M-codes:

M21: Tailstock Forward

M22: Tailstock Reverse

When an M21 is commanded, the tailstock quill moves forward and maintains continuous pressure. The tailstock body should be locked in place before commanding an M21.

When an M22 is commanded, the tailstock quill moves away from the workpiece. Hydraulic pressure is applied to retract the quill, then the hydraulic pressure is powered off. The hydraulic system has check valves that hold the position of the quill. Hydraulic pressure is then applied again on Cycle Start and on program looping M99 to ensure the quill remains retracted.

4.11.1 Tailstock Types

There are three basic types of tailstock: hydraulic quill, hydraulic positioned, and servo. The type of tailstock you have depends on the lathe model, and each type has different operation characteristics.

4.11.2 ST-10 Tailstock Operation

In the ST-10, you position the tailstock manually and activate a lock lever to hold it in place.



CAUTION:

Be sure to move the tailstock when necessary to avoid a collision.

The ST-10 tailstock has a fixed head, and a quill with 4" (102 mm) of travel. The only automatically moving part is the quill. Adjust the hydraulic pressure at the HPU to control quill hold force. Refer to the decal attached to the machine for information on quill hold force and hydraulic pressure.

You cannot move the tailstock quill with the **[HANDLE JOG]** control or the Remote Jog Handle. Also, **[POWER UP/RESTART]** or **[ZERO RETURN]** and **[ALL]** do not move the tailstock quill. The ST-10 tailstock does not have an axis assignment.

4.11.3 Hydraulic Tailstock (ST-20/30)

In ST-20 and ST-30 model lathes, a hydraulic cylinder positions the tailstock and applies hold force to the workpiece.

Adjust hydraulic pressure at the HPU to control tailstock hold force. Refer to the decal attached to your machine to determine the pressure setting for the hold force you need.

Recommended minimum hydraulic tailstock operating pressure is 120 psi. If hydraulic pressure is set lower than 120 psi, the tailstock may not function reliably.



NOTE:

During machine operation, [FEED HOLD] does not stop hydraulic tailstock motion. You must press [RESET] or [EMERGENCY STOP].

Hydraulic Tailstock (ST-20/30) Startup Procedure

If power to the lathe is shut off or interrupted while the hydraulic tailstock is engaged with a workpiece, the hold force is lost. Support the workpiece and zero return the tailstock to resume operation when power is restored.

4.11.4 ST-40 Servo Tailstock Operation

In ST-40 model lathes, a servo motor positions the tailstock and applies hold force to the workpiece.

Change Setting 241 to control servo tailstock hold force. Use a value between 1000 and 4500 pounds-force (if Setting 9 is INCH) or 4450 and 20110 Newtons (if Setting 9 is MM).

Tailstock load and current hold force are displayed as the B Axis in the axis load display (in modes such as **MDI** and **MEM**). The bar graph indicates current load, and the red line indicates the maximum hold force value specified in Setting 241. Actual hold force is displayed next to the bar graph. In **Jog** mode, this display appears in the **Active Tool** pane.

A hold icon [3] displays whether or not the tailstock is engaged. Refer to page 67 for more information on the tailstock hold icon.

ST-40 Servo Tailstock Startup Procedure

If power to the lathe is shut off or interrupted while the servo tailstock is engaged with a workpiece, the servo brake engages to preserve hold force and keep the tailstock in place.

When power is restored, the control displays the message *Tailstock Force Restored*. You can resume operating the lathe without zero returning the tailstock, if there are no **M22** commands in the program. These commands cause the tailstock to back away from the workpiece, which could then drop.

**CAUTION:**

Before you resume a program with an `M22` command after a power interruption, edit the program to remove or block delete the tailstock motion commands. You can then resume the program and complete the part. Keep in mind that until you zero return the tailstock, the control does not know the tailstock's position; therefore, Settings 93 and 94 do not protect the tailstock restricted zone from a crash.

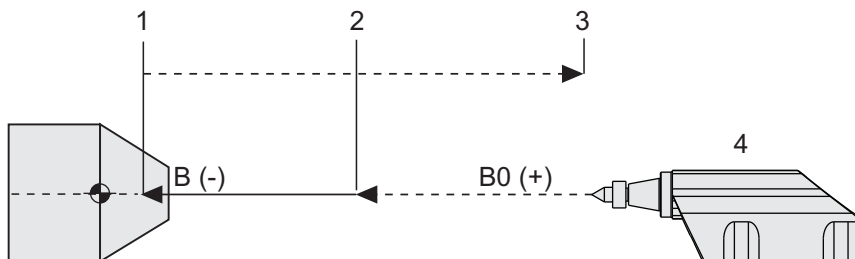
Zero return the tailstock before starting a new cycle on a new workpiece. You can then add the tailstock motion commands back into the program for future cycles.

The first use of the tailstock foot pedal after a power interruption zero returns the tailstock. Make sure the workpiece is supported before activating the tailstock foot pedal.

4.11.5 ST-20/30/40 Tailstock Operation

The ST-20/30/40 tailstock operation includes Settings, M-codes, Foot Pedal, and jogging functions.

F4.19: Setting 105 [3], 341 [2], 342[1], and [4] Home position.



4.11.6 Tailstock Settings

Below are the available tailstock settings:

- 93 - Tailstock X Clearance To learn more about this setting refer to page **384**
- 94 - Tailstock Z Clearance To learn more about this setting refer to page **384**
- 105 - Tailstock Retract Distance To learn more about this setting refer to page **386**
- 341 - Tailstock Rapid Position To learn more about this setting refer to page **405**
- 342 - Tailstock Advance Distance To learn more about this setting refer to page **405**

**NOTE:**

Settings 93, 94, 105, 341, and 342 do not apply to the ST-10 tailstock, since it is positioned manually.

4.11.7 Tailstock Foot Pedal Operation

When you press this pedal, the tailstock (or the tailstock quill) moves toward or away from the spindle, equivalent to an **M21** or **M22** command, depending on the current position. If the tailstock is away from the retract point, the foot pedal moves the tailstock toward the retract point (**M22**). If the tailstock is at the retract point, the foot pedal moves the tailstock toward the hold point (**M21**).

If you press the foot pedal while the tailstock is in motion, the tailstock stops and a new sequence must begin.

Press and hold the pedal for 5 seconds to retract the tailstock quill the full distance and maintain retract pressure. This makes sure the tailstock quill does not creep forward. Use this method to stow the tailstock quill any time it is not in use.



NOTE:

The tailstock position can change over time if it is left at a position that is not fully retracted or not in contact with a workpiece. This is due to normal hydraulic system leakage.

Use Setting 332 to enable or disable the tailstock pedal control. Refer to page **403** for more information.

4.11.8 Tailstock Restricted Zone

Setting up the tailstock involves setting a tailstock Restricted Zone.

Use Setting 93 and Setting 94 to make sure that the turret or any tools in the turret do not collide with the tailstock. Test the limits after you change these settings.

These settings make a restricted zone. The restricted zone is a protected rectangular area at the lower right of the lathe work space. The restricted zone changes so that the Z-Axis and tailstock maintain a safe distance from each other when below a specified X-Axis clearance plane.

Setting 93 specifies the X-Axis clearance plane and Setting 94 specifies the separation between the Z Axis and the B Axis (tailstock axis). If a programmed motion crosses the restricted zone, a warning message appears.

X Clearance Plane (Setting 93)

To set a value for the X clearance plane (Setting 93):

1. Place the control in **MDI** mode.
2. Select the longest tool that protrudes furthest on the X-axis plane in the turret.
3. Place the control in **Jog** mode.
4. Select the X-axis for jogging and move the X-axis clear of the tailstock.

5. Select the tailstock (B-axis) for jogging and move the tailstock beneath the selected tool.
6. Select the X-axis and approach the tailstock until the tool and tailstock are about 0.25" apart.
7. Back the tool away in the X Axis a small amount then enter the value for Setting 93.

Z- and B-Axis below the X Clearance Plane (Setting 94)

To set a separation for Z- and B-Axis below the X Clearance Plane (Setting 94):

1. Press **[ZERO RETURN]** and **[HOME G28]**.
2. Select the X-axis and move the turret in front of the tailstock quill tip.
3. Move the Z-axis so that the rear of the tool turret is within about 0.25" of the tailstock quill tip.
4. Enter the value in the Z-Axis **Machine Position** display for Setting 94.

Canceling a Restricted Zone

You may not always want to use a tailstock restricted zone (during setup, for example). To cancel a restricted zone:

1. Enter a 0 in Setting 94.
2. Enter the maximum X-Axis machine travel in Setting 93.

4.11.9 Jogging the Tailstock



CAUTION:

If you position the tailstock manually, do not use an M21 in your program. This makes the tailstock move away from the workpiece and then against the workpiece, which may cause the workpiece to drop. When a servo tailstock restores hold force after a power interruption, the tailstock is considered manually positioned since the control does not know the tailstock position until you zero return it.

You cannot jog the ST-40 servo tailstock while it is engaged with a workpiece, or while the spindle is running.

To jog the tailstock: